Reforming Composition Teaching Based on Artificial Intelligence: Model Innovation, Practical Bottlenecks, and Optimization Paths

Wei Wang[1], Minzhen Du[2]

[1] Zhejiang Normal University; Beijing Affiliated Elementary School of Chaoyang Normal School;P.R.China;wangweiblcu@163.com

[2] Zhejiang Normal University;P.R.China;315090494@qq.com

Abstract. With the rapid advancement of artificial intelligence technology, basic education has encountered unprecedented opportunities for digital transformation. In Chinese composition instruction, traditional teaching methods increasingly fall short of fostering students' expressive abilities and core competencies in the new era. Challenges such as insufficient writing motivation, significant disparities in expressive proficiency, and difficulties in delivering personalized guidance have become increasingly prominent. Leveraging robust data processing and intelligent feedback mechanisms, AI provides technical support for quantifying instructional objectives, diagnosing learning processes, and personalizing instruction, thereby shifting composition teaching from experience-based to scientific and precise approaches.A three-year school-based action research project, utilizing a "Chinese Language Digital Platform," has actively explored AI-driven writing instruction models and established an integrated teaching-learning-assessment system. Drawing on this practical experience, the paper reviews typical AI-assisted instructional approaches, analyzes implementation challenges, and proposes actionable optimization strategies. The study aims to provide theoretical insights and practical guidance for the digital transformation of Chinese writing instruction, foster comprehensive development of students' expressive abilities, and contribute to the high-quality advancement of Chinese language curriculum.

Keywords: Artificial Intelligence; Composition Teaching; Teaching Models; Practical Bottlenecks; Optimization Paths

1. Introduction

With the rapid advancement of artificial intelligence technology, the field of basic education has encountered unprecedented opportunities for digital transformation. Particularly in Chinese composition instruction, traditional teaching methods can no longer meet the comprehensive needs of fostering students' expressive abilities and core competencies in the new era. In recent years, challenges such as insufficient student motivation to write, significant disparities in expressive proficiency, and difficulties in providing personalized guidance have become increasingly prominent, necessitating systematic breakthroughs through emerging technologies. Leveraging its robust data processing capabilities and intelligent feedback mechanisms, AI technology provides technical support for quantifying instructional objectives, diagnosing learning processes, and delivering personalized instruction, thereby driving the shift of composition teaching from an experience-based approach to a more scientific and precise one.

Beijing Affiliated Elementary School of Chaoyang Normal School, leveraging the "Chinese Language Digital Platform," has actively explored innovative teaching models for writing instruction based on artificial intelligence and established an integrated teaching system encompassing teaching, learning, and assessment. Drawing on the school's practical experience, this paper reviews and analyzes typical AI-assisted writing instruction approaches, addresses challenges encountered in their implementation, and proposes actionable optimization strategies. The study aims to provide theoretical insights and practical guidance for the digital transformation of Chinese language writing instruction, foster the comprehensive development of students' expressive abilities, and contribute to the high-quality advancement of the Chinese language curriculum.

2.Innovations and practical foundations of writing instruction models based on artificial intelligence

2.1 The need for transforming writing instruction approaches under the guidance of core competencies

Writing is the most critical domain in Chinese language learning that reflects students 'linguistic competence, thinking qualities, and individual expression, serving as the core vehicle for implementing the "language construction and application" component of core literacy. However, writing instruction in frontline practice has long suffered from structural issues such as vague tasks, formalized training methods, fragmented guidance, and delayed evaluation. These problems result in weak student motivation to write, superficial content, limited linguistic diversity, inefficient teacher guidance, excessive instructional burden, and poor evaluation effectiveness, severely hindering the development of students' expressive abilities and the realization of Chinese language education's educational value.

With the new curriculum standards strengthening the learning task cluster of "Expression and Communication," writing instruction has evolved from traditional text reproduction and genre imitation to a more contextualized, task-oriented, and communicative approach. Teaching requirements have shifted from emphasizing "outcomes" to focusing on "processes," from prioritizing "content" to developing "competencies," and from concentrating on "output" to valuing "feedback." This presents greater challenges for teachers' instructional design, resource integration, and assessment capabilities, while also providing a practical pathway for integrating new technologies into the classroom.

2.2 Characteristics of the application of artificial intelligence technology in language teaching

With its capabilities in big data processing, natural language understanding, and intelligent feedback, artificial intelligence technology is becoming a pivotal force in reshaping educational paradigms and teaching methodologies. In the field of language education, AI technologies have been progressively implemented in areas such as speech recognition, automated grading, essay evaluation, writing guidance, and knowledge graph construction, demonstrating significant advantages in supporting composition instruction.

First, AI can overcome the temporal, spatial, and human resource constraints of traditional writing instruction, providing students with immediate, multidimensional, and personalized writing support. Second, AI technology effectively integrates and analyzes student writing data, facilitating a shift from "grading-based assessment" to "growth-oriented diagnosis," thereby making the development trajectory of writing skills visible and traceable. Third, AI enables precise delivery of teaching resources and dynamic generation of instructional tasks, enabling a closed-loop approach to writing instruction encompassing teaching, learning, and evaluation, which allows teachers to provide differentiated guidance and targeted interventions tailored to individual student needs.

2.3Analysis of a Typical AI-Enabled Writing Instruction Model

Through the deep integration of Beijing Affiliated Elementary School of Chaoyang Normal School with the "Chinese Language Digital Platform," the school has systematically explored a new paradigm for writing instruction supported by artificial intelligence, initially developing several representative teaching models:

2.3.1AI Diagnosis-Oriented Teaching Model

The AI diagnosis-driven teaching model stands as one of the most versatile and practical approaches for integrating artificial intelligence into composition instruction. It begins with students' preliminary writing drafts, utilizes AI-powered grading as an intermediary tool, employs data-driven teacher assessment, and ultimately delivers targeted personalized guidance along with meticulous revisions.

In this model, students complete initial drafts of their compositions based on unit themes before class and have them intelligently graded using AI writing tools such as the "Chinese Language Digital Assessment Platform," Doubao, and iFlytek. The system not only generates basic scores but also automatically analyzes common issues in essays—including structural flaws, expression biases, linguistic redundancy, and lack of thematic focus—to produce a "Writing Diagnostic Report" for students and a "Teaching Recommendation Set" for educators.

In their instructional design, teachers utilize AI-generated data to identify the most prevalent common challenges in students' writing and establish corresponding teaching objectives and training priorities. During classroom instruction, they focus not on covering every aspect but rather on high-frequency issues, thereby enhancing both the precision and effectiveness of practice sessions.

After class, students make personalized revisions to their original work under dual guidance from AI feedback and teacher recommendations, forming a closed-loop learning process of "initial drafting – diagnosis – practice – revision." The revised versions can be re-uploaded to the platform, where the system automatically updates the competency profile, enabling teachers to track improvement outcomes and support targeted instructional follow-up.

Taking the third-grade Chinese lesson "My Plant Friends" as an example, students completed a pre-class assignment on this topic and submitted it through the Chinese Language Platform to an AI grading system. The feedback indicated common issues such as disorganized writing sequences, lack of rhetorical devices, and insufficient imagination. Based on this, teachers precisely identified key teaching objectives: guiding students to describe events in chronological order, employ rhetorical techniques like metaphors, and develop creative imagination. During class, the teacher implemented a collaborative "Writing Circle" approach, having students work in roles to revise exemplary passages together, thereby mastering expressive strategies through joint revision. After class, students made secondary revisions to their original works incorporating both AI feedback and peer suggestions, achieving targeted writing practice and skill enhancement, thus establishing a closed-loop teaching process characterized by data-driven analysis, diagnostic instruction, and precise refinement.

2.3.2AI-Driven, Context-Based Teaching Model

The AI context-driven teaching model introduces anthropomorphic, task-oriented, and gamified intelligent characters to establish contextual interactions with students, naturally fostering writing skills through an engaging and enjoyable writing process. This approach is particularly suitable for lower and middle primary grades, focusing on the complete writing chain of "stimulating interest — supporting conception — guiding expression."

In the third-grade lesson "Those Distinctive People Around Us," the teacher utilized an AI assistant to create a personified character named "Game King," which guided students through writing challenges via voice or text prompts. The AI sequentially provided writing topics, guiding phrases, contextual questions, and structural suggestions, while acting as a "challenge instructor" to offer feedback and encouragement. For instance, Task 1 instructed students to "identify character traits"; Task 2 encouraged them to "highlight characteristics through a single event"; and Task 3 advised using rhetorical devices such as metaphors and personification. Through this engaging interactive process, students progressively developed ideas, accumulated knowledge, and expressed themselves, transforming their writing experience from passive completion to active engagement.

The greatest advantage of this model lies in its ability to significantly enhance students' enjoyment and immersion in writing, particularly for those with limited expressive skills or weak language organization abilities. The AI-driven contextual roles serve both as guides for writing and companions for expression. It overcomes the challenge of the "cold start" inherent in traditional composition teaching, enabling students to develop their language skills naturally in a relaxed and enjoyable environment.

2.3.3Human-Machine Collaborative Creation Model

The human-machine collaborative creation model represents a novel approach to writing instruction that emphasizes "joint authorship and dynamic generation." Its core principle involves leveraging AI technology to provide real-time assistance during students' writing process at stages such as idea development, expression, and revision, enabling them to broaden their perspectives, refine their expression, and enhance linguistic effectiveness while maintaining creative autonomy and authentic emotional expression. This model is primarily designed for upper secondary students, particularly those with foundational writing skills but lacking systematic writing strategies.

In this model, students independently determine writing topics, construct expressive frameworks, and gather linguistic materials around specific composition tasks, while the AI platform provides comprehensive support throughout the process—including keyword association assistance, structural template suggestions, and language style optimization guidance. For instance, in the fifth-grade composition lesson "Various Types of People," teachers utilize the "Digital Chinese Language Platform" to design a "Star Competition Task" that encourages students to employ AI tools for conceptual development. Before class, students complete character profile selection tasks; the platform automatically analyzes high-frequency vocabulary related to characters and provides personality labels along with example references. During class, teachers facilitate collaborative writing within a writing community, where AI tools identify textual highlights and logical inconsistencies in real time to assist revisions. After class, the platform generates multidimensional competency profiles, offering data-driven insights for instructional refinement and student self-improvement.

Artificial intelligence technology in writing instruction not only demonstrates technical advantages at the tool level but also drives profound transformations in teaching methodology, lesson design, and teacher-student relationships. The core purpose of platform empowerment lies not in replacing teachers, but in enhancing the effectiveness of teaching, the precision of learning, and the scientific rigor of assessment, thereby providing crucial support for achieving a high-quality, personalized, and ecosystem-driven transformation of writing education.

3. Analysis of Bottlenecks and Causes in AI-Based Composition Teaching Practice

Against the backdrop of artificial intelligence's extensive integration into Chinese language teaching in basic education, writing instruction in primary schools is undergoing a systematic paradigm shift from being "experience-driven" to "data-driven." Although diverse teaching models—such as "AI-diagnosis-oriented," "context-driven," and "human-machine collaborative creation-based" —have initially demonstrated potential for enhancing effectiveness and reducing instructional burdens in classroom practice, as educational reforms advance into more complex phases, deep-seated conflicts have emerged between technological tools, educational stakeholders, and pedagogical principles. Based on platform usage data and classroom observation records from Beijing Affiliated Elementary School of Chaoyang Normal School covering 32 classes and over 1,600 students over the past three years, this paper identifies three major bottlenecks hindering AI-enhanced writing instruction and examines their underlying structural causes.

3.1 The "Dual Constraints" Between Teachers' Digital Literacy and Teaching Philosophy: A Gap from "Operational Tools" to "Understanding Logic"

The integration of artificial intelligence into composition instruction is far more than a simple technological substitution; it represents a fundamental challenge to the professional knowledge and skill framework of Chinese language teachers. A key limitation lies in the fact that most educators remain at the "functional operation" stage when using AI tools—proficiently operating grading buttons, viewing score rankings, and accessing model essay templates—but lacking a basic understanding of the underlying principles of natural language processing (such as semantic similarity calculation, sentiment polarity analysis, and textual cohesion recognition). A pilot survey conducted at our institution during the fall semester of 2024 revealed that while 72% of Chinese language teachers could independently input commands into AI systems and export reports, only 18% could accurately explain how the "content richness" scores generated by these systems were derived from specific textual features (e.g., adjective density, number of examples, frequency of detailed descriptions).

A deeper constraint stems from the entrenched inertia in teaching philosophies. For years, the essay-teaching model of "defining learning based on instruction and substituting feedback for instructional guidance" has become deeply ingrained as a professional default. Teachers tend to focus their efforts solely on "pre-writing guidance" and "post-writing correction," neglecting the dynamic phase of "in-process facilitation" —the most data-driven aspect of the writing process. During classroom observations, we observed that when AI systems immediately identify a student's weakness in detailed description, many teachers' first reaction is not to adjust subsequent teaching strategies, but rather to use this diagnosis as grounds for ordering a rewrite. This essentially reinforces outdated pedagogical concepts through new technology, reducing data to an evaluation tool rather than a means for improvement.

This disconnect between technological empowerment and conceptual stagnation has created implicit cognitive resistance to the student-centered classroom transformation that AI should facilitate.

From a causal perspective, this constraint stems from two main aspects: First, in current teacher education and in-service training programs, the development of "Technology-Pedagogy-Content Integration" (TPACK) still emphasizes technical demonstrations over theoretical explanation, leaving teachers lacking the ability to act as a bridge between data and instructional decision-making; second, school management's evaluation focus has not shifted from outcome attainment to process optimization, resulting in a lack of institutional incentives and psychological security for teachers to experiment with new human-computer collaborative workflows.

3.2 The restructuring of instructional formats presents significant challenges: the lack of a structural framework for technological integration leads to fragmented effectiveness.

Within the current primary school Chinese language teaching system, "reading classes" have developed a relatively mature framework of lesson types, whereas "expression classes" (particularly writing instruction) have long remained characterized by vague lesson structures and arbitrary implementation processes. This structural deficiency has been further exacerbated when combined with AI tools. In our school's practice, AI technology is currently integrated into three key application scenarios: topic stimulation and material recommendation before writing, real-time linguistic refinement suggestions during writing, and grammar and structure analysis after writing. However, without a systematic lesson-type framework serving as a foundational framework, these applications can only function as temporary fixes rather than comprehensive solutions.

Observations reveal that the vast majority of teachers, after introducing AI, still adhere to a linear teaching approach: "model essays as guidance – imitation writing – focused feedback." AI tools are often integrated into peripheral stages such as "pre-class preparation" or "post-class revision," failing to truly become an integral part of the core teaching process—the complete writing journey involving student conception, drafting, revision, and proofreading—where they could serve as a readily accessible "cognitive partner." For example, in fifth-grade composition instruction on "writing about a memorable person," teachers pre-generated three model essays using AI for student reference, yet designed no interactive activities addressing "how to engage with AI to deepen character portrayal." The integration of AI essentially merely replaced the traditional role of exemplary essay collections without altering students' passive learning attitude.

The challenges in reconstructing teaching models stem from multiple underlying causes. Firstly, the inherent "non-linear and generative" nature of expression-based classes creates inherent tension with traditional classroom management models that emphasize standardized procedures; teachers commonly report difficulty in controlling class dynamics when AI is integrated. Secondly, most existing AI tools follow an assessment-oriented approach focused on error correction and scoring rather than a teaching methodology centered on inspiration and dialogue, and their user interfaces and output formats are inherently unsuitable for activities rooted in communication, sharing, and collaborative revision. Finally, school-based teaching research systems lack mechanisms for analyzing exemplary cases related to "AI-enhanced expression-based classes," leaving most teachers without reference points for replicating established models, which confines this reform primarily to individual educators' explorations and hinders its institutionalization and sustainable development.

3.3 Insufficient explanatory power of writing evaluation: "Black-box feedback" undermines the genuine validity of teaching improvement.

The depth and accuracy of writing assessment have long been a fundamental challenge in improving language education quality, yet the introduction of AI was expected to address this issue. However, feedback from systematic usage across grades three through six at our institution reveals that mainstream AI-based writing evaluation systems still exhibit significant limitations in functional comprehensiveness. While these systems have achieved considerable maturity in identifying superficial linguistic features such as grammatical errors, vocabulary richness, and sentence length variations, their assessment of deeper writing competencies — including logical coherence between paragraphs, authenticity of emotional expression, and originality with aesthetic value in language use—remains at a stage of probabilistic estimation. For instance, an essay using the template "Maternal love = offering an umbrella on a rainy night" often scores similarly high as a genuine creative piece based on real-life experiences in terms of "content richness" and "emotional intensity," with the former sometimes receiving higher scores due to more standardized wording. This precisely highlights structural shortcomings in current algorithms' ability to grasp the essence of authentic writing.

The more critical bottleneck lies in the "interpretative gap" in evaluation results. The system primarily outputs a set of quantitative indicators (e.g., "scored 85 points, good grade"), lacking specific explanations and pedagogical interpretations for deeper questions such as "why did you score 85 points?", "which descriptive elements enhanced the expressive effect?", or "what additional content is needed to strengthen the logic of which paragraph." When using this feedback, teachers often rely on personal experience to provide a secondary interpretation of the data, with the accuracy and consistency of such interpretations varying significantly; students, meanwhile, remain confused—they know what is happening but not why it happened. A fourth-grade student admitted in an interview: "I know my essay has 'an incomplete structure,' but I don't understand exactly where it's flawed or how to make it complete."

The direct consequence of this "black-box evaluation" approach is a dual erosion of validity: Firstly, personalized guidance has degenerated into formulaic recommendations — where the system automatically provides "improvement suggestions for low-scoring essays" or "model essays for high scorers" based on assessment results, representing essentially a standardized, context-free approach that falls far short of the goal of personalized instruction characterized by precision to specific student needs and key points; Secondly, teachers' trust in the AI system declined significantly after an initial period of enthusiasm. School tracking data shows that the frequency of teachers proactively using AI for in-depth instructional analysis peaked in the fourth month post-deployment, then declined monthly until dropping to 43% by the eighth month, with interviewed teachers consistently reporting that while the feedback was useful, it was insufficient.

4. Optimization Approaches and Ecosystem Development for Composition Teaching Based on Artificial Intelligence

To address the aforementioned practical challenges, this study proposes a systematic optimization framework for elementary school Chinese expression teaching in the context of artificial intelligence, structured around the logical thread of "goal integration – human-machine collaboration – ecosystem reconstruction." This framework aims to advance writing instruction from mere "integration of technological tools" toward profound transformations encompassing "reconstruction of teaching paradigms" and "reconstruction of learning ecosystems," with its core focus on achieving deep integration between technological logic and pedagogical logic.

4.1Capability Mapping-driven: Establishing a hierarchical and visualized framework for defining objectives

Traditional writing instruction has long faced challenges of vague objectives and fragmented competency progression, with a lack of comparable and measurable standards for skill advancement across educational stages, leading to an inefficient cycle of repetitive practice and experience-based teaching. The advent of artificial intelligence offers a breakthrough: through techniques such as semantic analysis and learning behavior data modeling, writing competence—a tacit literacy—can be transformed into a structured map that is both representable and diagnosable.

The research team developed a tiered indicator system encompassing seven competency dimensions— "problem analysis, topic selection, thematic development, structure construction, writing style, composition execution, and content sharing" — using the instructional requirements for the "Expression and Communication" domain in the Compulsory Education Chinese Curriculum Standards (2022 Edition) as a framework. This system integrates writing competency models from domestic and international writing psychology research, including Bereiter & Scardamalia's knowledge transfer model and Hayes & Flower's writing cognitive process theory. It provides operational definitions of cognitive levels and behavioral manifestations across lower, middle, and upper educational stages for each dimension, and employs natural language processing technology to convert qualitative descriptions into quantifiable metrics identifiable by the platform. Based on this, the platform generates a personalized "competency map" for each student, dynamically displaying their mastery levels, growth rates, and relative positions compared to grade-level norms, thereby supporting teachers in making precise instructional decisions.

In the sixth-grade "Writing Around Central Meaning" unit instruction at our school, the application of competency mapping transformed teaching objectives from vague perception to precise identification. Based on students' pre-writing diagnostic data, the platform automatically categorized them into three types— "idea-focused," "material-rich," and "structurally loose" —and assigned differentiated tasks accordingly. The experimental class ($N=42$) demonstrated significantly higher achievement in core competencies compared to the control class ($N=40$) ($t=3.21$, $p<0.01$), with the proportion of students reporting clarity about lesson content rising from 54% before curriculum reform to 89%. This indicates that competency mapping not only enables teachers to tailor instruction based on assessment results but also helps students develop clear self-awareness and learning direction.

4.2 Human-Machine Collaborative Teaching Model: Achieving Structured Reengineering of the Teaching Process

Simply introducing AI tools into traditional classrooms without altering the teaching structure often reduces technology to mere decoration rather than a driving force. The key to success lies in redesigning writing instruction processes according to the phased characteristics of the cognitive writing process, integrating AI as a cognitive partner throughout students' writing journey instead of merely serving as a grading tool for final outcomes.

This study builds upon Flower and Hayes's (1981) writing cognition model (planning –translation– modification), decomposing the writing instruction process into four functional lesson types: "topic analysis and material selection," "material construction," "writing refinement," and "creative expression." Each type corresponds to a specific stage of writing cognition, with clearly defined roles for AI across these categories. In the "topic analysis and material selection" session, AI facilitates "topic activation" and "material association," broadening students' thematic perspectives through semantic networks. During "material construction," AI employs "Socratic questioning" techniques to guide students in refining details and organizing logical structures of raw materials. In the "writing refinement" session, AI provides linguistic optimization suggestions such as vocabulary substitutions and sentence restructuring, which students evaluate and adopt selectively. For "creative expression," AI assumes a "reader" role, offering real-time feedback on textual novelty and aesthetic impact. Consequently, teachers' roles evolve from "lecturers" to "task designers," "process facilitators," and "deep dialogue partners" within each lesson type.

Our school implemented an experimental lesson format called "Material Construction Class" within the "Life Kaleidoscope" unit for fourth-grade students. The experimental class followed a standard workflow: AI generates micro-scenario prompt cards related to classroom contexts (30 per class); students engage in 2–3 rounds of interactive dialogue with the AI by clicking keywords, creating personalized material anchors; teachers then select representative cases from platform-collected data for targeted instructional guidance. After one semester of implementation, students 'in-class material output increased from an average of 87 characters to 156 characters—a 79.3% rise—with significantly enhanced personalization features in their writing (originality scores improved from 4.2/10 to 7.1/10 after independent blind evaluations by three Chinese language teachers). These results demonstrate that this lesson reform transformed AI's role from "external assistance" to genuine "internal integration," substantially enhancing students' cognitive engagement during the writing planning phase.

4.3 Multidimensional Evaluation Profile: Establishing a comprehensive and interpretable diagnostic feedback mechanism throughout the entire process

Writing assessment has long been plagued by two interrelated issues: first, the oversimplification of evaluation dimensions (overemphasizing linguistic accuracy while neglecting cognitive depth and individual expression); second, the lack of transparency in assessment outcomes (scores or grades fail to provide actionable teaching guidance). While current AI writing systems excel at grammatical correction and vocabulary analysis, they demonstrate limited explanatory power in evaluating deeper writing competencies. The solution lies in shifting the focus from outcome determination to process diagnosis, and enhancing the instructional interpretability of assessments through visual profiling mechanisms.

This study establishes a three-dimensional evaluation model comprising "process, outcome, and trend." The "process" dimension tracks students' behavioral data across writing stages (outline development, initial draft, revision, finalization), including modification frequency, types of revisions (vocabulary level, syntactic level, discourse level), and pause duration, to assess their metacognitive monitoring abilities. The "outcome" dimension incorporates three higher-order dimensions—logical coherence, emotional authenticity, and linguistic aesthetics — in addition to language standardization metrics, evaluated through human-machine collaborative annotation

(AI preliminary screening + teacher review). The "trend" dimension generates individual capacity development curves and predictive trajectories by longitudinally analyzing multiple writing samples. Based on these three-dimensional datasets, the platform creates a personalized "writing profile" for each student, presenting a radar chart that visualizes performance levels across five dimensions: expression intent, conceptualization ability, linguistic expression, revision capability, and logical construction. This profile automatically links to corresponding teaching resource libraries, providing teachers and students with a comprehensive feedback pathway encompassing diagnosis, attribution analysis, and strategy formulation.

A study enrolled 24 fifth-grade students whose writing proficiency ranked in the bottom 30% of their class for a semester-long portrait-based intervention. Analysis revealed that these students consistently exhibited weak performance in two key dimensions: "logical construction" (mean score: 58.2/100) and "revision skills" (mean score: 51.7/100). To address this, teachers implemented an intervention strategy combining "structural scaffolding" with "targeted revision guidance." The scaffolding component included specialized exercises using causal/qualitative connectors and structured paragraph templates, while revision guidance required students to focus on platform-marked structural issues during each revision session. After one semester, the mean scores improved significantly: logical construction reached 79.4 ($t=6.73$, $p<0.001$), and revision skills rose to 72.1 ($t=5.84$, $p<0.001$). More importantly, when writing independently without platform assistance later, students maintained these improved skills at a retention rate of 82% (as measured by post-intervention assessments), demonstrating both efficacy and transferability of this portrait-driven approach.

4.4 Platform Ecosystem Integration: Building a Data-Driven New Teaching and Research Community

The sustainability of AI-enhanced teaching does not depend on the sophistication of a single tool, but rather on its seamless integration with the school's curriculum framework, instructional research systems, and teachers' professional development pathways. If a platform remains merely an isolated technological system, its effectiveness will decline as teachers' initial enthusiasm wanes. Therefore, platform development must evolve from "tool deployment" to "ecosystem building."

The school advances ecosystem integration across both horizontal and vertical dimensions through its digital language education platform. Horizontally, it has established a comprehensive teaching resource system integrating five components: Task Library, Material Repository, Model Essay Collection, Assessment Database, and Case Study Archive. All resources are labeled according to competency frameworks, enabling intelligent matching between diagnostic outcomes, instructional objectives, and resource recommendations. Vertically, the platform implements a research-driven mechanism featuring data-driven analysis, problem-oriented approaches, collaborative task design, and co-developed resources: it automatically aggregates common writing data at grade and class levels (e.g., frequent error patterns and competency gaps) to inform instructional development; teaching teams conduct lesson studies and strategy collaborations based on identified issues; while teachers' instructional designs, reflective cases, and student exemplars are anonymized before being returned to the platform, fostering dynamic resource iteration and sharing. This ecosystem also integrates with regional quality assessment systems, allowing school-specific data to be interpreted within broader contextual frameworks.

4.5 Human-Machine Mutual Enhancement Competence: Cultivating Students' Autonomous Intelligent Writing Skills

The most frequently overlooked risk in AI-assisted writing education is that the convenience of technology may foster cognitive dependence among students, undermining their ability for independent expression and deep thinking. The key to mitigating this risk lies not in restricting technology use, but in consciously cultivating students' "digital literacy" —specifically, their comprehensive skills to understand tasks, critically evaluate suggestions, recognize linguistic aesthetic value, and make independent revisions within human-computer collaborative environments.

5. Conclusion

This study has established a three-tiered progressive literacy development framework within teaching practice. The first tier is the "Standardized Use" mechanism, which defines clear AI application boundaries for each writing task (e.g., green light encourages creative association with materials, yellow light restricts full-text polishing, red light prohibits direct text generation), helping students develop ethical use of technological tools. The second tier is the "Critical Dialogue" mechanism, featuring dedicated AI suggestion analysis sessions that guide students to compare AI-generated texts with original works in terms of language style, emotional authenticity, and personal distinctiveness, fostering critical evaluation habits toward AI outputs. The third tier is the "Originality Commitment" mechanism, requiring students to submit final drafts with a revision statement detailing adopted and rejected AI suggestions along with reasons, transforming the revision process from passive acceptance to active decision-making.

AI-powered writing instruction represents not merely an update to teaching methods, but a systematic restructuring of instructional logic, educational mechanisms, and curriculum ecosystems. The aforementioned five strategies address five key dimensions—goal setting, process reengineering, evaluation innovation, ecosystem integration, and literacy development—forming a mutually supportive closed-loop framework: the competency map provides cognitive benchmarks; human-machine collaborative lesson models serve as implementation vehicles; multidimensional profiles offer diagnostic foundations; platform ecosystems provide institutional safeguards; and literacy cultivation establishes fundamental foundations for learners. Through their synergy, these elements aim to elevate primary school Chinese writing instruction from a superficial level of technological assistance to a deeper focus on competency development, achieving an integrated, intelligent teaching paradigm encompassing "teaching–learning–evaluation–research."

References

[1] Bereiter, C., & Scardamalia, M. (1987). The psychology of written composition. Hillsdale, NJ: Lawrence Erlbaum Associates.

[2] Flower, L., & Hayes, J. R. (1981). A cognitive process theory of writing. College Composition and Communication, 32(4), 365–387.

[3] Weng Weibin, & Li Jinglai. (2020). 人工智能时代教育的应对 [Educational responses in the age of artificial intelligence] [J]. Theory and Practice of Education, 40(25), 3–7. [in Chinese]

[4] Li Yan, Jin Haoyue, Dong Chengli, et al. (2025). 基于生成式人工智能的中学生人机协同写作行为及影响因素研究 [Human-machine collaborative writing behaviors and influencing factors among secondary school students based on generative artificial intelligence] [J]. Modern Distance Education, (1), 22–33. [in Chinese]

[5] Liu Bangqi, Nie Xiaolin, Wang Shijin, et al. (2024). 生成式人工智能与未来教育形态重塑：技术框架、能力特征及应用趋势 [Generative artificial intelligence and the reshaping of future educational forms: Technical framework, capability characteristics, and application trends] [J]. E-education Research, 45(1), 13–20. [in Chinese]

[6] Xia Jingfang, & Lai Yiling. (2025). 生成式人工智能辅助教学的可能与路径——以中小学语文作文教学为例 [Possibilities and pathways of generative artificial intelligence-assisted instruction: A case study of Chinese composition teaching in primary and secondary schools] [J]. Journal of School Management, (2), 36–40. [in Chinese]

[7] Ministry of Education of the People's Republic of China. (2022). 义务教育语文课程标准（2022 年版） [Chinese language curriculum standards for compulsory education (2022 edition)] [S]. Beijing: Beijing Normal University Publishing Group. [in Chinese]

[8] Yu Ruili, & Han Zongpo. (2025). 智能批改赋能作文教学减负增效的循证实践——以中文作文智能批改系统应用为例 [Evidence-based practice of intelligent grading empowering composition teaching to reduce burden and increase efficiency: A case study of a Chinese composition intelligent grading system] [J]. Journal of Information Technology in Primary and Secondary Education, (1), 86–87. [in Chinese]